

Prevention and Treatment of Diabetic Kidney Disease

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INTRODUCTION

The development of diabetic kidney disease (DKD) signifies a generalized microvascular syndrome that is frequently accompanied by macrovascular disease (see [Chapter 31](#)). We use the terms DKD (and formerly diabetic nephropathy) to encompass the spectrum of vascular, glomerular, and tubulointerstitial components of chronic kidney disease (CKD) attributed to diabetes. Classically, DKD evolves through several stages based on urine albumin excretion and measured as urine albumin-to-creatinine ratio in a spot urine sample (UACR): normoalbuminuria (<30 mg/g), moderately increased albuminuria (previously called microalbuminuria, 30–300 mg/g), and severely increased albuminuria (previously called macroalbuminuria or overt nephropathy, >300 mg/g).¹ Furthermore, a substantial proportion of patients with diabetes mellitus have a nonproteinuric phenotype with progressive loss of estimated glomerular filtration rate (eGFR). DKD classification and staging are based on eGFR and degree of albuminuria, as in other causes of CKD. The level of UACR and eGFR are both independently associated with the risk of adverse kidney outcomes and cardiovascular (CV) morbidity and mortality. Normoalbuminuria is arbitrarily defined by UACR less than 30 mg/g, a threshold far above the normal albumin excretion in most healthy individuals. Thus, even in individuals with normoalbuminuria, increments in albuminuria are associated with increased cardiorenal risk.

In patients with established DKD, the goals of treatment are albuminuria regression and preservation of kidney function. Strict blood pressure (BP) and glycemic control early in the disease course are vital. Aggressive lipid-lowering and lifestyle modifications, including adherence to a low-protein and low-sodium diet, exercise, weight loss, and smoking cessation all are beneficial and likely to improve kidney and CV outcomes. Such multifactorial therapy lowered the risk for cardiovascular disease (CVD), nephropathy, retinopathy, and autonomic polyneuropathy in the Steno 2 trial in patients with type 2 diabetes (T2D), and even delayed mortality.² Most patients with advanced-stage DKD, however, are likely to die of CVD or progress to end-stage kidney disease (ESKD), even though treatment may slow progression.

This chapter reviews the management of patients with type 1 diabetes and kidney disease and focuses on strategies that promote kidney protection and cardioprotection in patients with T2D. The recent introduction of highly effective medications to ameliorate CV and kidney outcomes in patients with T2D and DKD has revolutionized the approach to clinical care, and management strategies are rapidly changing as new pharmaceuticals become available and as their indications and use in the lower eGFR ranges widen. [Fig. 32.1](#) is from the Kidney Disease: Improving Global Outcomes (KDIGO) DKD 2022 Guideline.^{2a} In general, the pyramidal peak describes the overall treatment goals, which are to optimize glycemic, blood pressure, and lipid control in patients with T2D. Treatment approaches begin with the foundational goals of smoking cessation and encouraging exercise. In addition, patients should be encouraged to eat a healthy diet, especially targeting lipid, sodium, and modest

protein restriction, encouraging fruits and vegetables, and achieving an optimal body weight. For patients with hypertension, hyperlipidemia, and CV disease complicating T2D, the KDIGO guidelines recommend first-line therapy to control glycemia with metformin (for patients with eGFR > 30 mL/min/1.73 m²); a sodium-glucose transporter inhibitor (SGLT2i, for patients with underlying CV and/or kidney involvement and eGFR > 20 mL/min/1.73 m²); blood pressure and proteinuria modulation with renin-angiotensin system (RAS) inhibition or angiotensin receptor blockade; and control of hyperlipidemia with a statin. For those in need of additional or alternative medication approaches to achieve glycemic, blood pressure, proteinuria, CV, and/or lipid goals, second-line therapies including glucagon-like peptide 1 (GLP-1) receptor agonists may be used to address hyperglycemia, proteinuria, renal function decline, and CV complications. A nonsteroidal aldosterone antagonist may be added to further reduce proteinuria, slow the rate of loss of eGFR, and reduce CV complications. Finally, antiplatelet agents may be used to address atherosclerotic CV disease.

One obstacle to achieving adherence is the complexity of these regimens. Therefore, prevention and treatment of patients with DKD must be individualized and require consideration of the cost, side effects, and convenience of the drug regimen measured against the anticipated benefits. A comprehensive approach incorporating multidisciplinary teams, along with patient participation in shared decision-making, is best for managing patients with DKD. Special considerations are indicated in the management of the diabetic patient with advanced CKD (see [Chapter 33](#)). Many therapeutic issues discussed here are not specific for DKD and may be relevant for CKD in general (see [Chapter 82](#)). The paragraphs below expand on specific considerations in using particular medications or medication classes.

DIABETIC KIDNEY DISEASE IN TYPE 1 DIABETES

In patients with T1D, strict glycemic control decreases the risk for developing DKD. The Diabetes Control and Complications Trial (DCCT) compared the effects of intensive glucose control with conventional treatment on the long-term complications of T1D ([Fig. 32.2](#)). Over 9 years, patients with mean hemoglobin A1c (HbA1c) of 7% who received intensive therapy had a 35% to 45% lower risk for development of moderately increased albuminuria compared with the control group (mean HbA1c, 9%).³ In the DCCT follow-up, the Epidemiology of Diabetes Interventions and Complications (EDIC) trial, the long-term risk for developing impaired glomerular filtration rate (GFR) was 50% lower in patients assigned to tight control compared with those treated with conventional therapy. This effect was not evident until more than 10 years after randomization.⁴

The appearance of moderately increased albuminuria typically precedes hypertension in patients with T1D. When hypertension occurs, control is imperative. Higher BP is associated with increasing

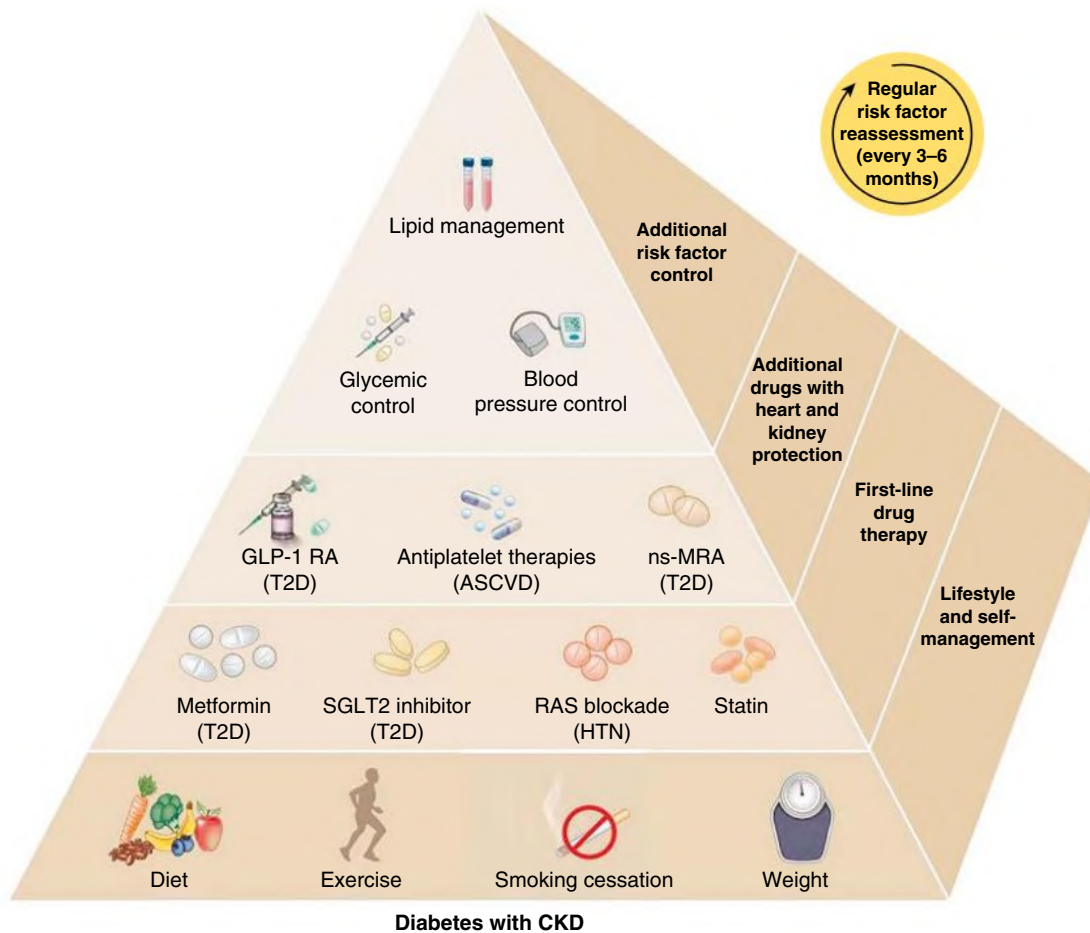


Fig. 32.1 Treatment Algorithm for Diabetic Kidney Disease (DKD) in Patients With Type 2 Diabetes (T2D). The Kidney Disease: Improving Global Outcomes (KDIGO) Executive Summary describes a recommended approach for the management of DKD, beginning with foundational lifestyle recommendations. Built upon these are the sequential incorporation of first- and second-line glycemic lowering medications, with preferential consideration of each drug's potential for cardioprotection and renoprotection above and beyond their capacity to improve glycemic control. At the top of the pyramid are overarching goals of risk factor modification, including optimization of glycemic, hypertensive, and lipid management. ASCVD, Atherosclerotic cardiovascular disease; HTN, hypertension; SGLT2, sodium-glucose transporter 2. (From Rossing P, Caramori ML, Chan JCN, et al. Kidney Disease: Improving Global Outcomes [KDIGO] Executive Summary of the KDIGO 2022 Clinical Practice Guideline for diabetes management in chronic kidney disease. An update based on rapidly emerging new evidence. *Kidney Int.* 2022;102:990–999.)

albuminuria, more rapid progression, and increased risk for kidney failure, as well as increased risk for fatal and nonfatal CV events.⁵ Thus, effective treatment of hypertension is arguably the single most important strategy in the treatment of established DKD (Fig. 32.3). Antihypertensive therapies, regardless of the agent used, reduce UACR, delay progression of nephropathy, postpone loss of GFR, and improve survival.

Lower rates of ESKD among patients with T1D attributable to BP and tight glycemic control, along with antiproteinuric strategies (see section on RAAS blockade), are now reported from many countries around the world, and ESKD, when it does occur, occurs at an older age.⁶ These encouraging data underscore the need for early glycemic and BP control in T1D.

DIABETIC KIDNEY DISEASE IN TYPE 2 DIABETES

Glycemic Control

Several major studies demonstrate a lower risk for nephropathy with stricter glycemic control in patients with T2D. In a design similar to the DCCT, the Kumamoto study found a 60% reduction in moderately increased albuminuria in relatively young, nonobese patients with

T2D receiving intensive glycemic treatment (HbA1c 7.1%) compared with conventional treatment (HbA1c 9.4%).⁷ In the United Kingdom Prospective Diabetes Study (UKPDS) trial, newly diagnosed patients with T2D were assigned to intensive management (HbA1c 7.0%) with a sulfonylurea or insulin or to conventional management (HbA1c 7.9%) with diet alone. After 9 years of intensive therapy, the risk of developing moderately increased albuminuria was reduced by 24%.⁸ After termination of the study, patients were observed for a further 10 years. The differences in HbA1c were lost within 1 year, but a 24% lower relative risk of microvascular disease and myocardial infarction (–15%) persisted. All-cause mortality was also reduced by 13%. This phenomenon of ongoing beneficial effects on diabetic complications after a period of improved glycemic control even if followed by a return to less intensive metabolic control has been termed *metabolic memory* or *legacy effect*. It emphasizes the importance of early glycemic control in the primary prevention of microvascular and macrovascular complications in diabetic patients.

Most of the evidence favoring strict glycemic control comes from studies of patients with normoalbuminuria or early stages of DKD. Fewer studies addressed intensive glycemic control in patients with more advanced DKD stages, in whom it may be difficult to show

Intensive Glucose Control Reduces Development of Moderately Increased Albuminuria

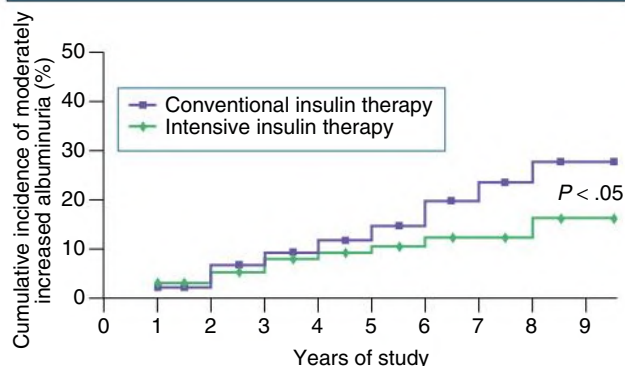


Fig. 32.2 Moderately Increased Albuminuria Risk Reduction With Intensive Versus Conventional Insulin Therapy. Intensive glucose control was associated with a decreased risk for the subsequent development of moderately increased albuminuria in patients with type 1 diabetes. (Modified from Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *N Engl J Med.* 1993;329:977–986.)

Control of Blood Pressure Slows Progression of Type 1 Diabetic Kidney Disease

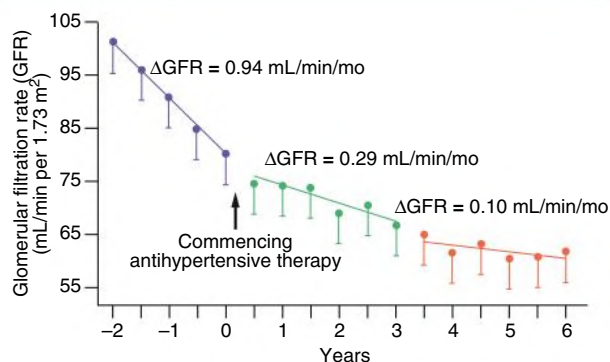


Fig. 32.3 Control of blood pressure reduces the risk for progression in type 1 diabetes and diabetic kidney disease. (Modified from Parving HH, Andersen AR, Smidt VM, et al. Effect of antihypertensive treatment on kidney function in diabetic nephropathy. *Br Med J (Clin Res Ed).* 1987;294[6585]:1443–1447.)

benefit because the results are confounded by the effects of concomitant hypertension and CVD. Even so, there is evidence to support glycemic control in reducing the risk for worsening albuminuria and kidney functional decline.

Glycemic Targets

HbA1c is widely used to monitor glycemic control, and testing should be performed routinely in all patients with diabetes as part of continuing care, combined with patient self-monitoring of blood glucose or continuous glucose monitoring in selected patients. The HbA1c assay has limitations in patients with advanced CKD, as the result can be influenced by conditions that affect red cell turnover (see [Chapter 33](#)).

Several major trials tested whether strict glycemic targets reduced CVD risk in patients with T2D. The Action in Diabetes and Vascular Disease, Perindopril and Indapamide Controlled Evaluation

(ADVANCE) study showed that intensive glycemic control (HbA1c 6.5% vs. 7.3%) yielded a 10% relative reduction in major macro- and microvascular events, in particular a 21% relative reduction in nephropathy.⁹ However, in the Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial, very tight glycemic control (lowering HbA1c to a median of 6.4% vs. 7.5% with conventional control) led to a 22% increase in mortality from any cause and did not significantly reduce major CV events.¹⁰ A third major study of tight glucose control in patients with T2D, the Veterans Affairs Diabetes Trial (VADT), found no significant reduction in CV deaths or events over 7.5 years in high-risk patients treated aggressively for glycemic control (median HbA1c 6.9%) compared with standard therapy (median HbA1c 8.4%).¹¹ A meta-analysis showed that tight glycemic control reduced microvascular complications, including kidney and eye events.¹²

We interpret the data to suggest that glycemic control must be individualized, considering the patient's age, duration of diabetes, presence of CVD, presence of CKD, and microvascular risks and complications, as well as previous glycemic control and susceptibility to and awareness of hypoglycemia. In younger patients with short diabetes duration, longer life expectancy, low hypoglycemia risk, and no prior CV events, strict glycemic control can reduce DKD risk and other microvascular complications. A more cautious approach to glycemic control is recommended in the frail or elderly patient with long-standing diabetes or preexisting CV problems or in patients who are susceptible to hypoglycemic episodes. In these patients, the HbA1c goal may be set higher. The Kidney Disease: Improving Global Outcomes (KDIGO) 2020 clinical practice guidelines recommend lowering HbA1c to a goal ranging from less than 6.5% to less than 8.0%.¹³ The American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD) similarly endorse tight glycemic control at the onset of diabetes in patients with few comorbidities as a preventive strategy and less stringent control in patients with more advanced comorbidities.¹⁴

Antihyperglycemic Therapeutic Options in Type 2 Diabetes and Kidney Disease

Options for antihyperglycemic treatment in patients with T2D have expanded over the past decade, and several confer CV and kidney protection substantially beyond their glycemic effects. Despite specific indications and relative contraindications for some newer drug classes discussed below, most patients can safely be prescribed at least one, if not more, classes of the newer antiglycemic agents as needed. These newer agents have been shown to confer a level of CV and kidney protection that far surpasses the protection conferred by RAS inhibition alone. Importantly, the severity of CKD has implications for the efficacy and safety of glycemic treatments. Dosing adjustments and careful monitoring for adverse effects are required for patients with advanced CKD or with kidney failure (see [Chapter 33](#)). [Fig. 32.4](#) represents a treatment strategy combining lifestyle recommendations and a hierarchical medication prescribing approach recommended in the KDIGO DKD Guideline^{2a} to address glycemic control and CV and renoprotection in patients with T2D. The following paragraphs expand on the particular pharmacologic considerations of medication use in each of the drug classes.

Metformin

Metformin is a biguanide which has been used for over 60 years and is the preferred initial antihyperglycemic drug for T2D because of its low cost, high efficacy, and low risk for hypoglycemia. Metformin also has weight- and lipid-lowering properties, and beneficial effects on CV and all-cause mortality.¹⁵ It is primarily eliminated by the kidney, and the risk of lactic acidosis in patients with kidney impairment or acute illness is a concern. However, the lowest eGFR safety threshold for metformin has been reconsidered in recent years, as data demonstrating an increased risk of lactic acidosis in CKD is limited. Currently, it is

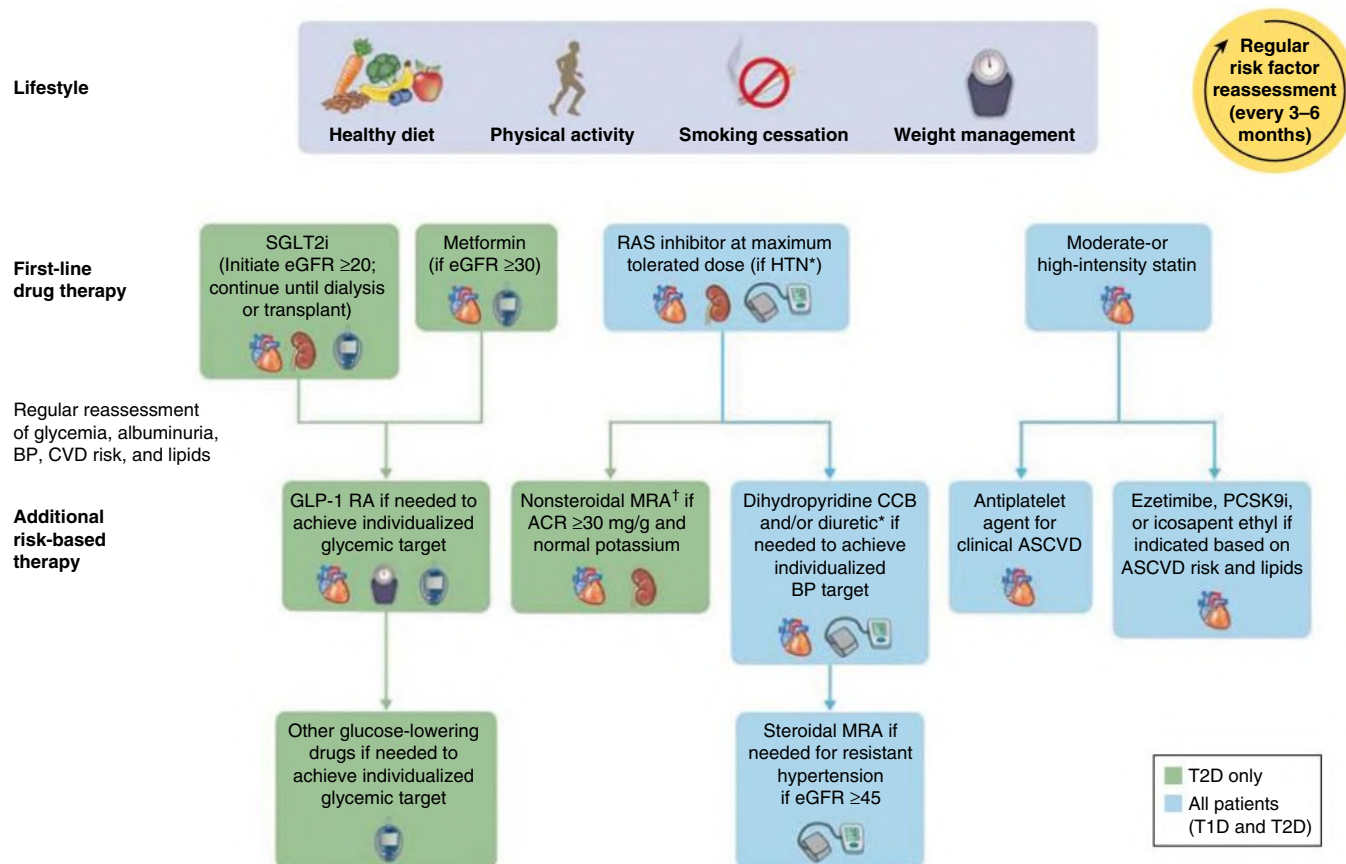


Fig. 32.4 Treatment algorithm for glycemic control in patients with type 2 diabetes and diabetic kidney disease. ACR, Albumin/creatinine ratio; ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CCB, calcium channel blocker; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HTN, hypertension; MRA, mineralocorticoid receptor antagonist; PCSK9i, proprotein convertase subtilisin/kexin type 9 inhibitor; RA, receptor agonist; RAS, renin-angiotensin system; SGLT2i, sodium-glucose cotransporter-2 inhibitor; T2D, type 2 diabetes. (From Rossing P, Caramori ML, Chan JCN, et al. Executive summary of the KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease: an update based on rapidly emerging new evidence. *Kidney Int.* 2022;102:990-999.)

recommended to decrease the dose of metformin to 500 mg twice daily at eGFR less than 45 mL/min/1.73 m² and discontinue use at eGFR less than 30 mL/min/1.73 m².^{2a}

Sodium Glucose Cotransporter-2 Inhibitors

The magnitude of the effect size of SGLT2i for kidney protection in T2D is approximately 2.5 times the protective effect of RAS inhibitor when added to a maximally tolerated dose of RAS inhibitor. No drug class currently exists with this degree of renoprotection. Empagliflozin, dapagliflozin, canagliflozin, and ertugliflozin are now widely approved antihyperglycemic therapies with kidney benefits conferred at least in part via glycosuric and tubuloglomerular feedback mechanisms. They induce osmotic diuresis, which in turn confers natriuretic effects contributing to plasma volume contraction, have decreased weight, and have decreased systolic BP by 4 to 6 mm Hg and diastolic BP by 1 to 2 mm Hg. SGLT2 inhibition is associated with an acute reduction of more than 10% in eGFR in about a third of all individuals with T2D and CVD. However, this initial decline is fully reversible after stopping SGLT inhibition. Albuminuria is reduced by 30% to 40%. These effects mirror preclinical observations suggesting that proximal tubular natriuresis activates renal tubuloglomerular feedback through increased macula densa sodium and chloride delivery, leading to afferent vasoconstriction. Glycosuric effects are attenuated in patients with

eGFR less than 60 mL/min/1.73 m², but BP, eGFR, and albuminuria lowering effects are preserved.

There is solid evidence for CV and kidney protection of SGLT2 inhibitors in patients with T2D. The Empagliflozin Cardiovascular Outcome Event (EMPA-REG OUTCOME) trial in patients with T2D and established CVD reported a 14% reduction in the primary composite outcome of CV death, nonfatal myocardial infarction, nonfatal stroke, and more than 30% reductions in CV mortality, overall mortality, and heart failure hospitalizations.¹⁶ The Canagliflozin Cardiovascular Assessment Study (CANVAS) showed a similar magnitude of CV event reduction using canagliflozin in patients with T2D and a high risk of CVD. In terms of kidney protection, the EMPA-REG OUTCOME study reported a 39% reduction in incident or worsening kidney disease that included doubling of serum creatinine (relative risk reduction, 44%) and kidney-replacement therapy (relative risk reduction, 55%) in the empagliflozin group, whereas the CANVAS-Renal trial similarly reported an impressive 40% reduction in the composite kidney outcome (defined as a sustained 40% reduction in the rate of eGFR decline, need for kidney replacement therapy, or death from kidney causes).^{17,18} The Canagliflozin and Renal Events in Diabetes with Established Nephropathy Clinical Evaluation (CREDENCE) trial was halted early after a planned interim analysis indicated significant 30% reduction in composite outcome (ESKD, doubling of the serum creatinine, or death from kidney or CV causes) in the canagliflozin group

compared with the placebo group.¹⁹ Similarly, dapagliflozin was shown to reduce heart failure and fatal myocardial outcomes in patients with T2D and CV risk factors in the DECLARE study.^{19a} and the DAPA-HF study.^{19b} Of great significance, the DAPA-CKD study showed that treatment with dapagliflozin improved all of the components of the composite kidney endpoint of sustained eGFR decline, ESKD, and CV or renal death in patients with and without DKD with a relative risk reduction of 39%.²⁰ In 2022, similar findings of CV and renoprotection were reported for empagliflozin^{20a} in patients with and without T2D. Corroborating all of the results of these individual studies, a meta-analysis^{20b} reported that SGLT2 inhibitors reduced the risk of kidney disease progression by 37%, the risk of acute kidney injury by 23%, and the risk of cardiovascular death or hospitalization for heart failure by 23%, with similar effects in those with and without diabetes. SGLT2 inhibitors also reduced the CV death risk but did not reduce the risk of non-CV death. The authors concluded that the absolute effects of SGLT2 inhibitors outweighed the small risks of ketoacidosis or amputation, the latter having been observed in only one trial with canagliflozin. The risk for genital and urinary tract infections, especially with *Candida* species, is generally increased, and hygiene measures are recommended.²⁰

Collective evidence from clinical studies demonstrated that SGLT2 inhibitor use does not predispose to clinically significant AKI.

GLP-1 Receptor Agonists

GLP-1 receptor agonists are a newer class of antihyperglycemic treatment with demonstrated CV and kidney benefits in patients with T2D. They lower glucose levels and reduce body weight and BP. The Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results (LEADER) trial showed that fewer patients in the liraglutide group died from CV causes (relative risk reduction, 22%), and the rate of death from any cause was also lower (15%). Fewer patients reached a kidney endpoint (23%), which consisted mainly of a lower frequency of heavy albuminuria.²¹ Similarly impressive CV benefits were demonstrated for semaglutide in the SUSTAIN-6 and PIONEER-6 studies.²² Both the 2020 ADA guideline and the 2022 KDIGO guideline now recommend the use of GLP-1 receptor agonists with proven CV benefits in patients with T2D who have established CVD or are at high risk. Notably, the 2020 ADA guidelines recommend using these agents for the reduction of CV risk independent of glycemic control,¹⁴ and the 2022 KDIGO guidelines recommend these as adjunctive therapies after consideration of metformin and SGLT2 inhibitors.^{2a} Cost concerns and the need for injection remain barriers to using GLP-1 receptor agonists. Additional research is needed to fully characterize subgroups among patients with T2D who will particularly benefit from GLP-1 receptor agonists and to determine if the CV and kidney benefits extend to patients with T1D.

Dipeptidyl Peptidase-4 Inhibitors

Like the GLP-1RA, the dipeptidyl peptidase-4 (DPP-4) inhibitors (i.e., gliptins) are incretin-based antihyperglycemic agents effective in lowering blood glucose in patients with T2D. By inhibiting DPP-4, a regulator of GLP-1 degradation, these agents increase GLP-1 levels. Thus, combined use of DPP-4 inhibitors and GLP-1 RA is not recommended. As a class, DPP-4 inhibitors are generally well tolerated and can be used even in patients with advanced CKD. Treatment with DPP-4 inhibitors reduces albuminuria. So far, clinical trials of DPP-4 inhibitors have demonstrated no additional CV outcome benefit compared with placebo.²³

Blood Pressure Control

As many as 40% of patients with T2D have known hypertension before the diagnosis of DKD. Once severely increased albuminuria develops, hypertension is almost universal and is associated with volume expansion and salt sensitivity. The absence of hypertension in an untreated

patient with T2D and DKD should raise suspicion for underlying cardiac disease.

The optimal BP target in DKD remains unclear. Guidelines published before the Action to Control Cardiovascular Risk in Diabetes Blood Pressure (ACCORD BP) trial suggested a BP target in diabetic patients of less than 130/80 mm Hg. However, this BP target was challenged by the ACCORD BP trial. Among patients with T2D and high CV risk, randomized to goal systolic BP of under 120 mm Hg or standard therapy aiming for under 140 mm Hg, there was no difference in the risks for composite major CV events. The risks for hyperkalemia and kidney dysfunction were increased at lower BP goals in ACCORD.²⁴ In contrast, the large Systolic Blood Pressure Intervention Trial (SPRINT) showed that for nondiabetic patients, lowering systolic BP below 120 mm Hg, compared with the standard goal below 140 mm Hg, significantly lowered event rates and mortality without a significant increase in adverse events.²⁵ Unfortunately, data are more limited in patients with diabetes or advanced stage CKD. In the Irbesartan Diabetic Nephropathy Trial (IDNT), progressive lowering of systolic BP to 120 mm Hg was associated with improved kidney and patient survival, an effect independent of baseline kidney function.²⁶ Mortality increased with systolic BP below 120 mm Hg, although a cause-and-effect relationship could not be inferred. Low diastolic BP is poorly tolerated, and the incidence of myocardial infarction and mortality increases at values less than 70 mm Hg, at least in patients with coronary heart disease, presumably because coronary perfusion occurs during diastole. Indeed, in the IDNT study, CV mortality increased not only with higher systolic BP but also with low diastolic BP.

The KDIGO 2021 Hypertension Guideline remained somewhat agnostic about the blood pressure target in patients with diabetes. Citing SPRINT, the workgroup report stated “The working group feels that cardiovascular benefits of intensive BP lowering cannot be excluded in patients with concomitant diabetes and CKD and that a large randomized trial addressing this issue is warranted.”²⁷ Thus, a solid evidence-based recommendation for a BP target in patients with diabetes remains elusive, but based on KDIGO 2021, it appears appropriate to target a BP less than 140/90 mm Hg for all diabetic patients and less than 130/80 mm Hg as tolerated for patients with CKD and/or UACR greater than 30 mg/g.²⁷ The American College of Cardiology/American Heart Association (ACC/AHA) 2017 hypertension guideline similarly set a BP target below 130/80 mm Hg for individuals with CKD and others at elevated CV risk. For diabetic patients at highest risk for cerebrovascular accidents, lower systolic BP goals (i.e., <120 mm Hg) may provide greater protection against stroke, but the potential risks for adverse events and the burden of antihypertensive therapy must be considered.

RAAS BLOCKADE

Renin-Angiotensin-Aldosterone System Inhibition in the Prevention of Diabetic Kidney Disease

We do not recommend the use of renin-angiotensin system (RAS) inhibition in normotensive, normoalbuminuric diabetic patients for the primary prevention of DKD. Most patients with diabetes do not develop DKD, even after long periods of uncontrolled hyperglycemia, and there are hazards using RAS inhibitors, including their potential teratogenicity. In a post hoc analysis of the multicenter Diabetic Retinopathy Candesartan Trials (DIRECT) program, which included patients with normotensive and normoalbuminuric T1D and normoalbuminuric T2D with or without hypertension, candesartan had no effect on the development of moderately increased albuminuria.²⁸

In hypertensive, normoalbuminuric patients, an angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) are both effective as first-line antihypertensive agents. The

Bergamo Nephrologic Diabetes Complications Trial (BENEDICT), which randomized patients with hypertension, normoalbuminuria, and T2D to placebo, verapamil, trandolapril, or a combination of verapamil plus trandolapril, showed less progression to moderately increased albuminuria in those receiving trandolapril either alone or with verapamil.²⁹ Results with verapamil alone were similar to placebo. There were similar findings in smaller studies with other RAS inhibitors, implicating a class effect. Longer-term studies would be required to demonstrate the effects of RAS inhibitors on the clinically important outcomes of death, dialysis, and doubling of serum creatinine level in normoalbuminuric patients.

Renin-Angiotensin System Inhibition in the Treatment of Diabetic Kidney Disease

In patients with established DKD, RAS inhibition confers kidney protection that is independent of BP reduction. The mitigation of intraglomerular hemodynamic and nonhemodynamic kidney effects of angiotensin II best explain the observed protection. Supporting this hypothesis, *in vitro* models of DKD show cellular effects of RAS inhibition that are consistent with benefits independent of BP effects (see Chapter 31).

Type 1 Diabetes

In patients with T1D with moderately increased albuminuria, ACE inhibitors reduce the risk for progression to overt nephropathy. In a meta-analysis in normotensive patients with T1D and moderately increased albuminuria treated with ACE inhibitors, the majority for more than 2 years, treatment was associated with a 60% reduction in progression to severely increased albuminuria and a threefold increase in regression to normoalbuminuria.³⁰ Changes in BP do not entirely explain the antiproteinuric effect of ACE inhibition.

In patients with severely increased albuminuria or overt nephropathy, captopril reduced albuminuria, slowed loss of GFR, and delayed the onset of kidney failure compared with placebo.³¹ The beneficial effect of captopril was greater in patients with reduced GFR at baseline largely because one of the components of the composite endpoint, a doubling of baseline serum creatinine level, was achieved more quickly in these patients.

Data are insufficient to demonstrate the efficacy of ARBs in T1D and DKD. Nevertheless, based on the shared properties of ACE inhibitors and ARBs in inhibiting the RAS, ARBs are probably effective in the treatment of T1D and DKD.

Type 2 Diabetes

In patients with T2D, more data are available on the kidney-protective effect of ARBs compared with ACE inhibitors. In the stage of moderately increased albuminuria, the IRMA 2 study showed that irbesartan reduced progression to overt nephropathy by 70% in patients with hypertension and T2D during a 2-year follow-up period.³² In the MARVAL trial, valsartan produced a greater reduction in UACR than did amlodipine (44% vs. 8%), with the same degree of BP reduction, suggesting that the antiproteinuric effect of ARBs is BP independent.

In patients with T2D with severely increased albuminuria and decreased GFR, the large randomized controlled trials (RCTs) IDNT and RENAAL showed that ARBs are effective in lowering proteinuria and decreasing the relative risk for reaching the composite endpoint of death, dialysis, and doubling of serum creatinine level.^{33,34} However, the risk reduction for reaching the composite endpoint was only 18% to 20% in these studies in patients with T2D and nephropathy, compared with the more robust risk reduction of about 50% in patients with T1D receiving captopril. ARBs did not decrease CV death in these trials but did decrease the incidence of heart failure.

Compared with ARBs, data on the efficacy of ACE inhibitors in T2D and DKD are less strong, largely because of small sample size or short follow-up. In general, ACE inhibitors and ARBs have similar effects on patient outcomes. In a small RCT of patients with T2D with early DKD and 5-year follow-up, telmisartan was not inferior to enalapril in providing long-term kidney protection.³⁵

Aldosterone Blockade in Diabetic Kidney Disease

The beneficial effect of ACE inhibitors and ARBs in slowing progressive kidney disease did not differentiate between the relative contributions of the RAS inhibition versus aldosterone system blockade (see Chapter 82). In fact, plasma aldosterone levels are elevated in a subset of patients despite ACE inhibitor and ARB therapy (also known as aldosterone breakthrough; see Chapter 82). In studies that defined aldosterone breakthrough as any increase from an individual's baseline serum aldosterone level (i.e., before RAS inhibition), the incidence ranged from 40% over 10 months to 53% over 12 months.³⁶ In addition to its classic effects of promoting sodium retention and enhancing potassium and magnesium excretion, aldosterone promotes tissue inflammation and fibrosis. Small studies demonstrated faster decline in GFR in patients who experienced aldosterone breakthrough (median, -5.0 mL/min/y) than in those who did not (median, -2.4 mL/min/y). However, current evidence is not strong enough to support widespread testing for aldosterone breakthrough. In selected patients, the addition of a mineralocorticoid receptor antagonist (MRA) may represent optimal therapy for patients who no longer show maximal antiproteinuric effects with RAS inhibitors.

The steroidal MRAs spironolactone and eplerenone reduce proteinuria when used alone. They have an additive effect on proteinuria when combined with an ACE inhibitor or ARB.³⁷ However, the risk for hyperkalemia, AKI, and other adverse effects, including gynecomastia, erectile dysfunction, and dysmenorrhea, frequently limit the use of combined MRA with ACE inhibitors or ARBs, especially in patients with reduced eGFR. Finerenone, a new nonsteroidal MRA, has greater receptor selectivity and affinity compared with steroidal MRAs (e.g., spironolactone and eplerenone) and has been approved by the US Food and Drug Administration and the European Medicines Agency for the treatment of patients with T2D to reduce CV morbidity and mortality and slow the rate of progressive renal functional decline. Recently published results from the two phase III finerenone studies, FIGARO and FIDELIO, demonstrated that finerenone reduced the relative risks of CKD progression and CV morbidity and mortality in patients with T2D CV risk and CKD.^{38,38a} Adverse events were similar between the finerenone and placebo group. Hyperkalemia leading to finerenone discontinuation was uncommon (2.3%) compared with placebo (0.9%). The success of finerenone to confer CV and kidney benefits and a low incidence of underlies the recommendation in the KDIGO DKD Guidelines to consider utilizing finerenone in a triple combination drug therapy strategy of ACE inhibitors or ARBs, SGLT2 inhibitors, and the nonsteroidal MRAs to provide optimal CV and kidney protection for most patients with T2D and DKD. Trials of several additional novel MRAs (esaxerenone, apararenone, KBP-5074) are underway. Recently, a phase III study of esaxerenone in patients with T2D showed albuminuria reduction compared to placebo.⁵³ Esaxerenone is approved in Japan for the treatment of hypertension.

Combination Therapy With Renin-Angiotensin System Inhibitors

Combination therapy with an ACE inhibitor and an ARB is not recommended for DKD based on the results of several large trials in which the combination failed to improve clinical outcomes but which did

engender hyperkalemia and/or AKI.³⁹ Similar untoward events were observed when an ACE inhibitor or ARB was combined with the direct renin inhibitor aliskerin.⁴⁰

Dosing and Adverse Effects Associated With ACE Inhibitors and ARBs

In individual patients, proteinuria may respond to ACE inhibitor or ARB dose escalations beyond those recommended for BP control.⁴¹ Unfortunately, maximal dosing of ACE inhibitors or ARBs may be limited by side effects, including hyperkalemia, hypotension, and AKI. In women of reproductive age, counseling about pregnancy prevention and contraceptive use should begin before a RAS inhibitor is started.

Serum creatinine concentration may increase up to 30% in proteinuric patients with kidney impairment after starting a RAS inhibitor. This rise in creatinine is associated with long-term kidney protection, and therapy should not necessarily be stopped in these patients. Increases in serum creatinine concentration above 30% after initiation of a RAS inhibitor should raise the suspicion of renal artery stenosis. Aggressive dose increments of ACE inhibitors or ARBs, especially in conjunction with diuresis, can precipitate AKI, so caution is appropriate. Interestingly, a retrospective cohort study showed that discontinuing RAS inhibition in patients with eGFR less than 30 mL/min/1.73 m² was associated with 39% and 37% higher risks for all-cause mortality and major adverse cardiovascular events (MACE), respectively, over 5 years of follow-up compared with continuation.⁴² No significant difference in the risk of progression to ESKD was found. A randomized clinical trial of RAS inhibition withdrawal in patients with advanced kidney disease (STOP-ACE) is currently underway. In practice, the addition of diuretic therapy and potassium binders can reduce the incidence of hyperkalemia in patients with severe CKD and offer the potential to enhance the therapeutic effects of RAS inhibitors.

TREATMENT OF DYSLIPIDEMIA

Most patients with DKD have dyslipidemia characterized by low levels of high-density lipoprotein (HDL) cholesterol, high triglyceride (TG) levels, and a shift from larger toward smaller low-density lipoprotein (LDL) cholesterol. Dyslipidemia in diabetic patients may contribute to the development of glomerulosclerosis and progressive kidney disease.⁴³ In the Diabetes Atherosclerosis Intervention Study (DAIS), patients with T2D taking fenofibrate had a significantly lower rate of progression from normoalbuminuria to moderately increased albuminuria at 3 years compared with the placebo group.⁴⁴

Statins have the strongest evidence for reducing atherosclerotic CVD risk and are the first-choice hypolipidemic agent. Treatment emphasis is placed on assessing a patient's global risk for CVD and using maximum tolerated statin intensity for primary and secondary prevention of CVD. For diabetic patients ages 40 to 75 years without atherosclerotic CVD, the ACC/AHA guidelines recommend moderate-intensity statin therapy. Younger patients with elevated CVD risk factors may reasonably be treated with a statin as well. In diabetic patients with atherosclerotic CVD, maximally tolerated dose of high-intensity statin is recommended, aiming to lower LDL cholesterol by 50% or more.⁴⁵ Follow up fasting lipid measurements should be done 4 to 12 weeks after starting a statin and should be repeated every 3 to 12 months as needed. The addition of a nonstatin hypolipidemic agent (e.g., ezetimibe) may be considered in high-risk patients who have not achieved a LDL cholesterol reduction goal on statin therapy. Once a patient has ESKD and is placed on dialysis, statin therapy may not improve CV outcomes (see Chapter 33).⁴⁶

In patients with T2D, LDL cholesterol is not the sole lipoprotein that defines CV risk. As major statin trials have demonstrated, lowering

of LDL cholesterol does not prevent the majority of adverse CV events and does not bring the CV risk in T2D down to the level of nondiabetic patients (referred to as residual CV risk). Atherogenic dyslipidemias, specifically elevated TG, low HDL cholesterol, elevated apolipoprotein B, and elevated apolipoprotein C III, are thought to be key factors associated with residual CV risk in diabetic patients.⁴⁷ In the UKPDS, elevated TG was independently associated with albuminuria in patients with T2D. Thus, interventions aimed at improving all lipid targets are recommended. However, it is not clear whether this is best achieved by intensification of statin therapy or supplementation of statin therapy with other hypolipidemic therapeutics or omega-3 fatty acids. In the ACCORD Lipid trial, routine use of combination therapy with a statin plus a fibrate did not reduce CV risk in patients with T2D.⁴⁸

NONPHARMACOLOGIC INTERVENTIONS

For all patients with diabetes mellitus, emphasis should be placed on lifestyle modification to lower the risk for DKD and CV events, including dietary restriction of salt and saturated fat, weight reduction and exercise as appropriate, and smoking cessation.

Dietary protein restriction may alleviate uremic symptoms in patients at or approaching ESKD. Small trials have shown low-protein diets (≤ 0.8 g/kg/day) reduce proteinuria significantly with increased plasma albumin in patients with T2D with severely increased albuminuria. A recent meta-analysis concluded that a low-protein diet (≤ 0.8 g/kg/day) slows the decline in eGFR and improves proteinuria in patients with DKD.⁴⁹ Even patients with mild DKD appear to benefit from dietary protein restriction with decreased proteinuria. However, nutritional counseling is advised for all patients to avoid protein-energy malnutrition and to receive education on salt, potassium, and phosphate restriction, as well as choice of carbohydrates and fats (see Chapter 90). The efficacy of low-protein diets in reducing the rate of kidney function loss in patients taking SGLT2 inhibitors has not been systematically studied.

Smoking cessation and weight reduction can provide additive kidney benefits and lower the risk for CV events in patients with DKD. Smoking is an independent risk factor for the development of nephropathy in T2D and is associated with accelerated loss of kidney function. Smoking cessation attenuates albuminuria and improves kidney prognosis.⁵⁰ Weight reduction also may improve kidney outcome. The Look-AHEAD trial found intensive lifestyle interventions consisting of diet and exercise plans achieved greater weight loss compared with diabetes support and education (mean 1-year weight loss of 8.6% vs. 0.7%) and reduced the incidence of CKD (0.63% vs. 0.91%) in obese patients with T2D.⁵¹ Furthermore, results of prospective cohort studies and emerging evidence from randomized clinical trials show bariatric surgery may prevent or slow the progression of DKD over medical intervention in obese patients with T2D.⁵²

EMERGING TREATMENTS FOR DIABETIC KIDNEY DISEASE

Genetic and epigenetic factors, glomerular hypertension and hyperfiltration, overactive RAS, accumulation of advanced glycation end products, oxidative stress, kidney inflammation, and fibrosis contribute to DKD and are potential therapeutic targets.

A number of treatments, some experimental and others in clinical use for other indications, have been tried to prevent or treat DKD. Unfortunately, many investigational drugs failed to meet primary clinical endpoints, had unacceptable side effects, or were abandoned by pharmaceutical companies for financial reasons. Bardoxolone methyl

TABLE 32.1 Selected Investigational Drugs for Diabetic Kidney Disease

Drug	Mechanism of Action	Clinical Trials and Comments
Finerenone, esaxerenone, aparenone, KBP-5074	Selective nonsteroidal MRAs	In FIDELIO-DKD, finerenone reduced kidney progression and CV events in T2D with advanced CKD. ³⁸ A phase III trial of esaxerenone in T2D with moderately elevated albuminuria showed that it is effective in reducing albuminuria. Esaxerenone is approved in Japan for the treatment of hypertension.
Atrasentan, avosentan	Endothelin A receptor blocker	In a phase III trial (SONAR), atrasentan reduced kidney endpoints by 35% in patients with T2D; significant risk for fluid retention and anemia occurred in the atrasentan group. ⁵⁵ A phase III trial of avosentan (ASCEND) showed a reduction in albuminuria in the short term, but the trial was terminated early due to fluid retention in the avosentan group. ⁵⁴
Paricalcitol	Vitamin D receptor activators	An RCT ($n = 281$) found addition of paricalcitol to RAS inhibition nonsignificantly lowered albuminuria in patients with T2D and DKD. ⁵⁷
Allopurinol, febuxostat, topiroxostat	Purine analog and inhibitor of xanthine oxidase that decreases uric acid formation	The PERL and CKD-FIX trials concluded that allopurinol did not reduce the progression of DKD. ^{58,59}
Pentoxifylline, CTP-499	Nonselective phosphodiesterase inhibitors	In a small trial ($n = 169$) of patients with T2D, the addition of pentoxifylline to a RAS inhibitor had additive antiproteinuric effect and slowed kidney disease progression after the first year of treatment and maintained statistical significance with placebo after 24 months. ⁶⁰
CCX-140, NOX-E36	C-C chemokine receptor 2 antagonist and chemokine ligand 2 inhibitor	Phase II trials of several chemokine inhibitors have shown kidney protective effects of chemokine inhibitors in T2D with DKD.
Selonsertib (GS-4997)	Selective ASK1 inhibitor	A phase II trial of selonsertib showed no effect on albuminuria compared with placebo, though post hoc analyses suggest that it may slow DKD progression. ⁵⁶
ASP-8232	Vascular adhesion protein 1 inhibitor	Ongoing phase II trial to evaluate the efficacy and safety of ASP-8232 in patients with T2D and DKD.
Bardoxolone methyl (RTA-402)	Inducer of Nrf2 pathway	A phase III trial of bardoxolone methyl in DKD was halted because of excess serious adverse events in the treated group. A separate phase II study in Japan showed significant improvement in eGFR, with an associated increase in UACR, in the bardoxolone methyl group compared with placebo. There is an ongoing phase III trial in Japanese patients with T2D and CKD.

ASCEND, Avosentan on Doubling of Serum Creatinine, ESKD and Death; ASK1, apoptosis signal-regulating kinase 1; CKD, chronic kidney disease; CV, cardiovascular; DKD, diabetic kidney disease; eGFR, estimated glomerular filtration rate; FIDELIO-DKD, Finerenone in Reducing Kidney Failure and Disease Progression in Diabetic Kidney Disease; MRA, mineralocorticoid receptor antagonist; RAS, renin-angiotensin system; RCT, randomized controlled trial; SONAR, Atrasentan and Renal Events in Patients with Type 2 Diabetes and Chronic Kidney Disease; T1D, type 1 diabetes; T2D, type 2 diabetes; UACR, urine albumin-to-creatinine ratio.

(an inducer of the Nrf2 pathway), ruboxistaurin (a protein kinase C inhibitor), pirfenidone (an antifibrotic agent), pimagidine/aminoguanidine/pyridoxamine (advanced glycation end product formation inhibitors), baricitinib (janus kinase inhibitor), palosuran (a vasopeptidase inhibitor), probucol (an antioxidant), and sulodexide (a glycosaminoglycan) are examples. Ongoing trials exploring new DKD treatments include endothelin receptor antagonists, drugs that target oxidative stress, xanthine oxidase inhibitors, and other antiinflammatory and antifibrotic agents (Table 32.1).

ENDOTHELIN RECEPTOR ANTAGONISTS

The kidney endothelin system is activated in experimental and clinical DKD. The antiproteinuric effect of an endothelin A receptor blocker was shown in the Avosentan on Doubling of Serum Creatinine, ESKD and Death (ASCEND) phase III clinical trial. However, fluid retention led to early termination of the ASCEND trial.⁵⁴ Another endothelin A receptor blocker, atrasentan, is more selective than avosentan and was shown in the Atrasentan and Renal Events in Patients with Type 2 Diabetes and Chronic Kidney Disease (SONAR) trial to reduce the incidence of primary composite endpoints of serum creatinine doubling or onset of ESKD. Serious adverse events included fluid retention and anemia, which occurred more frequently in the atrasentan group.

There were no CV benefits.⁵⁵ Whether atrasentan will be approved for DKD treatment remains unclear.

SELECTIVE APOPTOSIS SIGNAL-REGULATING KINASE 1 INHIBITOR

Apoptosis signal-regulating kinase 1 (ASK1) activation from oxidative stress in glomerular and tubular cells has been implicated in the pathogenesis of DKD. Selonsertib is a selective ASK1 inhibitor developed as a once-daily oral agent for the treatment of DKD. In a phase II trial evaluating selonsertib's safety and efficacy, selonsertib failed to reduce UACR compared with placebo, though post hoc analyses suggest that it may slow DKD progression.⁵⁶ A phase IIb trial enrolling patients with moderate-advanced DKD is currently underway.

Vitamin D Receptor Activators

Vitamin D receptor activation reduced albuminuria in small trials of DKD. A cross-sectional analysis of the National Health and Nutrition Examination Survey (NHANES, 2001–2006) data showed that there was an independent association between vitamin D deficiency and insufficiency with the presence of nephropathy. The largest phase II RCT investigating the effect of paricalcitol (a vitamin D analog) 1 or 2 µg/day in patients with T2D failed to meet its primary endpoint

(change in albuminuria), but post hoc analysis showed lowered albuminuria compared with placebo in patients with high sodium intake and on a higher dose of paricalcitol.⁵⁷ However, only 58% of the patients assigned to 2 µg/day of paricalcitol received the full dose during the study, and the follow-up period of 24 weeks was brief. The use of vitamin D analogs in clinical applications of DKD demands further RCTs in larger patient series with adequate follow-up.

Xanthine Oxidase Inhibitors

Epidemiologic studies suggest an independent association between asymptomatic hyperuricemia and increased risk for arterial hypertension, CKD, albuminuria, CV events, and mortality. Whereas some smaller clinical trials have shown improvements in BP control and slowing of CKD progression using urate-lowering therapy, two recent RCTs did not. In one trial treating patients with CKD at high risk of progression, urate-lowering treatment with allopurinol did not slow the decline in eGFR compared with placebo.⁵⁸ Similarly, in another trial, clinically meaningful benefits of serum urate reduction with allopurinol on kidney outcomes among patients with T1D and early to moderate DKD could not be detected.⁵⁹ Although both studies had some weaknesses in design (e.g., including subjects who were not hyperuricemic), the use of allopurinol for this indication is currently unsupported.

Phosphodiesterase Inhibitors

Pentoxifylline (PTF) is a nonselective phosphodiesterase (PDE) inhibitor used for symptomatic relief of claudication. It has

vasodilatory and antihypertensive effects and may have beneficial kidney hemodynamic effects. PTF also inhibits the synthesis of inflammatory cytokines. In a small trial in patients with T2D, the addition of PTF to RAS inhibition had additive antiproteinuric effects and slowed kidney disease progression after the first year of treatment. Statistical significance was maintained for 24 months.⁶⁰ A key problem for the clinical development of PTF as a DKD treatment is the low cost of the drug. A more selective PDE inhibitor, PF-489791, was discontinued despite early promising phase II results.

Novel Therapeutic Approaches

Novel investigational agents, including CCX-140, NOX-E36, DMX-200 (chemokine inhibitors), LY3016859 (an epidermal growth factor receptor inhibitor), GKT137831 (a NADPH oxidase inhibitor), ASP-8232 (a vascular adhesion protein 1 inhibitor), SER150 (a thromboxane A2 receptor antagonist), and several drugs currently in clinical use for other indications (e.g., N-acetylcysteine, colchicine), are being evaluated for treatment of proteinuric kidney disease, including DKD. A new formulation of bardoxolone methyl, RTA 402 (an inducer of Nrf2 pathway), is being tested for safety and efficacy in Japanese patients with T2D with CKD. The role of micro-RNAs in the pathogenesis of DKD is an emerging field and may also provide additional novel treatment approaches. New insights into the molecular mechanisms that underlie the origin and progression of DKD are emerging from large-scale genetic and molecular studies in experimental models and humans.

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